Quang Nguyen

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Education

Georgia Institute of Technology Atlanta, GA
M.S. in Electrical and Computer Engineering *Expected May 2028*
University of Rochester Rochester, NY
B.S. in Electrical and Computer Engineering, Minor in Computer Science; GPA: 3.8/4.0 *May 2026*

Experience

Undergraduate Research Assistant Rochester, NY
University of Rochester *May 2025 – May 2026*

- Built an LTspice electrothermal model of superconducting nanostrip single-photon detectors from material properties, dimensions and electrical characteristics to simulate the electrical response of multi-photon behaviors.
- Designed a photon-number-resolver in PSCAN2 using DC-to-SFQ converters, splitters, and coincidence correlators (AND cells with adjustable time-window) that converts each detector pulse to produce corresponding number of SFQ pulses.

Project Assistant Rochester, NY
Laboratory for Laser Energetics *Feb 2025 – May 2025*

- Conducted pump-probe spectroscopy with femtosecond pulses to measure carrier relaxations in GaAs samples, producing the response data for material characterization.
- Aligned the optics, synchronized the pulse trains, and tuned measurement parameters for the strongest time-resolved signal, working towards MOKE measurements with a rotating two-magnet field.

Hardware Engineer Intern Ho Chi Minh City, Vietnam
Vita Imaging *May 2024 – July 2024*

- Designed a USB-UART serial board for a medical device, using FT232RL and LDO regulator to support 1.8/2.8 V rails. Used ferrite beads and TVS diodes for a ESD protection link.
- Created the system block diagram and schematics in OrCAD Capture and laid out the PCB in Allegro, applying design rules for trace width and differential-pair spacing, leading to a working prototype with USB 2.0.

Projects

MIPS Pipelined Processor | *Verilog, Verilator* Dec 2025

- Designed a 32-bit MIPS processor as a five-stage pipeline with separate instruction and data memory, a 32-entry register file, ALU, branch unit, and control unit, running compiled MIPS programs in Verilator simulation.
- Built hazard-detection and forwarding units that clear most data hazards without stalling, insert one bubble on load-use hazards, and resolve branches early in the decode stage to cut control-hazard flushes.

Headphone Amplifier | *HSPICE* May 2025

- Designed a two-stage headphone amplifier with an OPA1656 preamp into a BUF634A output buffer, delivering 100 mW into 32 Ω with 0.72% THD for high-fidelity audio.
- Simulated stage gain, bandwidth, and input/output impedance in HSPICE. Soldered the board and compared measurements with waveform analysis in Python and MATLAB.

DC Motor Control | *PIC32, ADC/PWM* May 2024

- Implemented closed-loop motor-control program in C on a PIC32MX795F512L microcontroller, using a 48-CPR encoder through a 99:1 gearbox for speed feedback.
- Controlled the motor with a dual MC33926 H-bridge at 20 kHz PWM, setting speed and direction from a potentiometer via the ADC and tuning the control loop over UART with MATLAB.

Technical Skills

Languages:	Verilog, C/C++, Python, MATLAB, Java, MIPS Assembly, Lisp, Git Bash.
Software:	LTspice, HSPICE, OrCAD Capture, Allegro PCB, KiCad, Quartus, MPLAB X, PSCAN2.
Hardware:	Oscilloscope, Multimeter, Waveform Generator, Soldering.
Courses:	Electronic Circuits, IC Design, Digital Logic Design, Semiconductor Devices, Computer Org, Embedded Systems, Signal Systems.